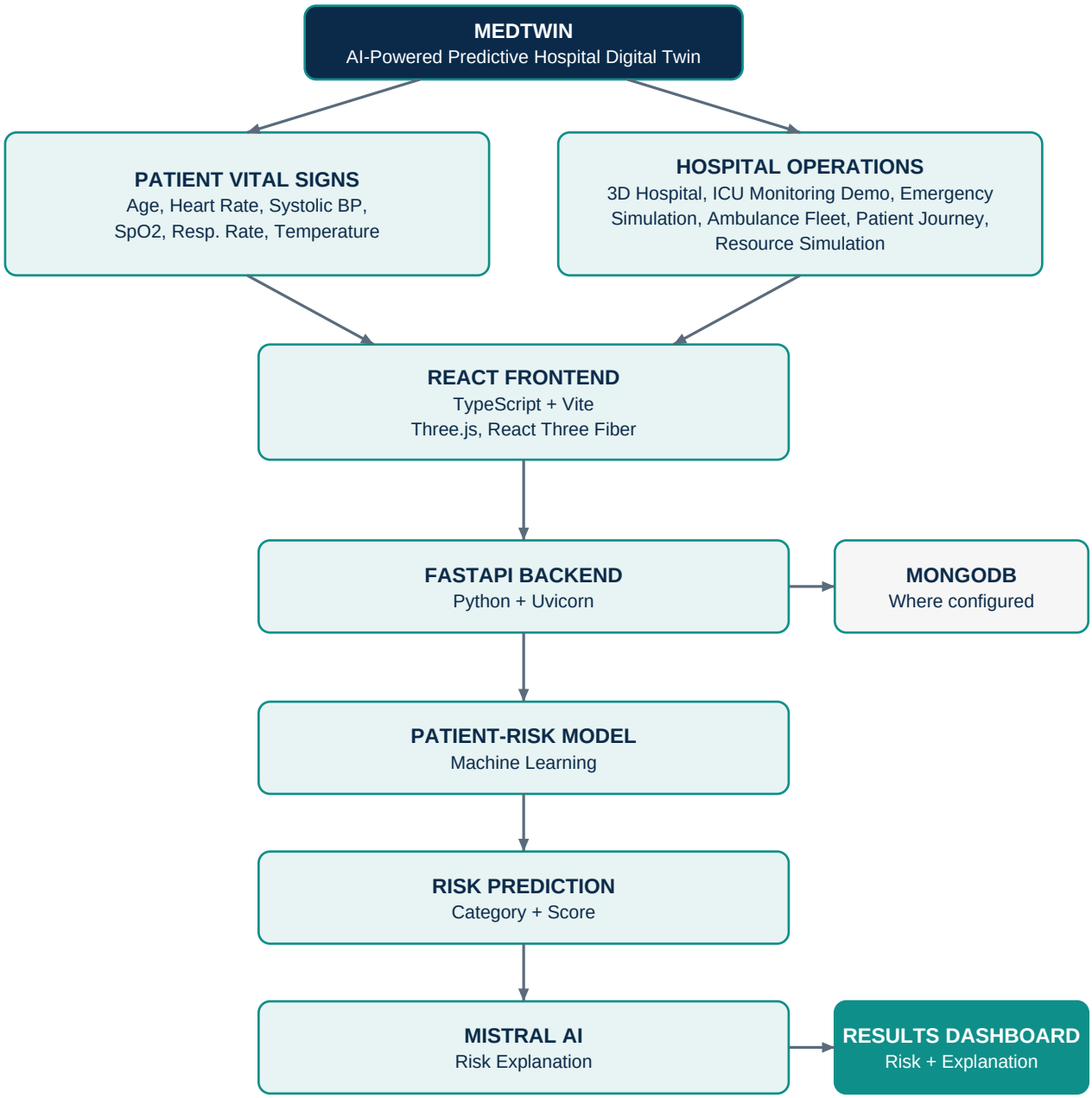
MedTwin: System Architecture

MedTwin combines patient vital-sign risk prediction with a 3D hospital operations digital twin. Patient data flows from the React frontend to a FastAPI backend, through a machine learning model, and Mistral AI turns the result into a human-readable explanation.



Workflow: Patient Risk Prediction

1	Patient enters vital signs
2	React Frontend
3	FastAPI Backend
4	Machine Learning Model
5	Risk Prediction
6	Risk Level (LOW / MODERATE / HIGH) and Risk Score or Probability
7	Mistral AI
8	Human-Readable Explanation
9	React Dashboard

Data flows downward. Each stage passes its output to the next.

Technology Stack

Component	Technology
Frontend	React, TypeScript, Vite
3D Visualization	Three.js, React Three Fiber
Backend	Python, FastAPI, Uvicorn
Patient-Risk Prediction	Machine Learning
AI Explanation	Mistral AI (mistral-small-latest)
Database	MongoDB, subject to implementation verification

Database: MongoDB, where configured and used by the application.